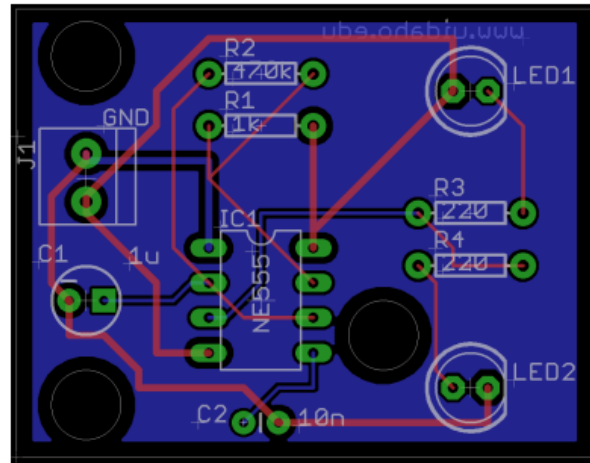
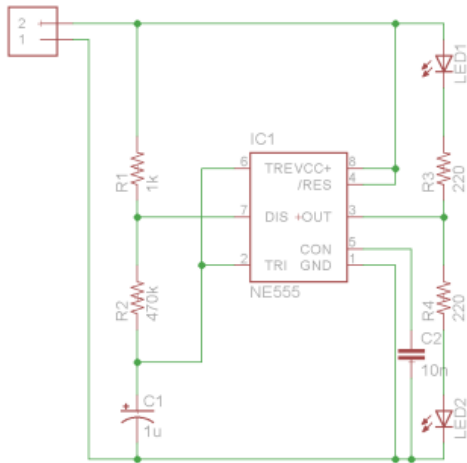


2023-2024
 3rd year /Electronic III
 Electronic and Telecommunication Department.
 Engineering Science/ Kufa University

Electronic Design Lab course

Electronic Design Lab 02: Design LED flash with 555 timer



Process steps

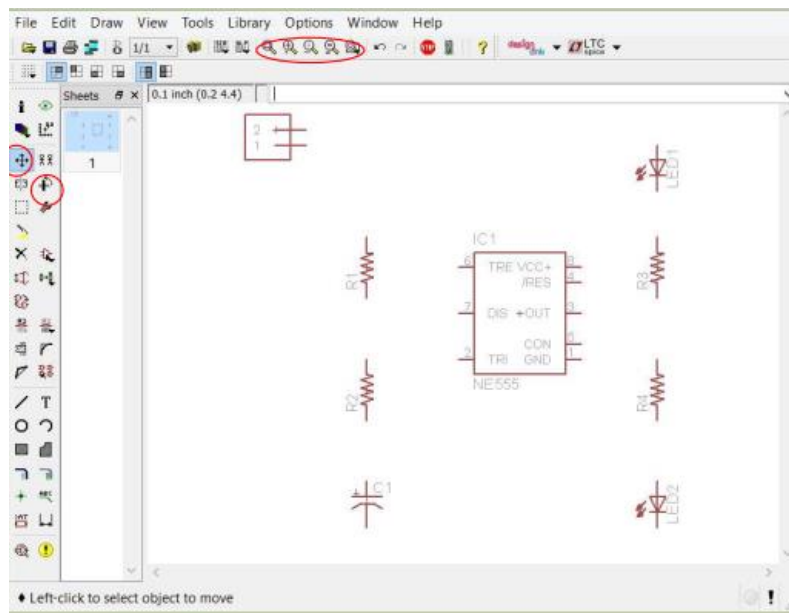
1. Create and name a new project.
 File->New->Project (you can name the folder as flasher)
2. Creating a New Schematic
 - Right click folder “flasher”->New->Schematic
3. Save and Rename of New Schematic
 - Schematic File->Save as... (flasher.sch).
4. Finding and Add a Component
 - Enter “*555*” in the Search field
 - Find “NE555”.
5. Add Components – Resistors and Capacitor
 - SparkFun-Resistors: RESISTORAXIAL-0.3 (x4)

- SparkFun-Capacitors: CAP_POLPTH2.

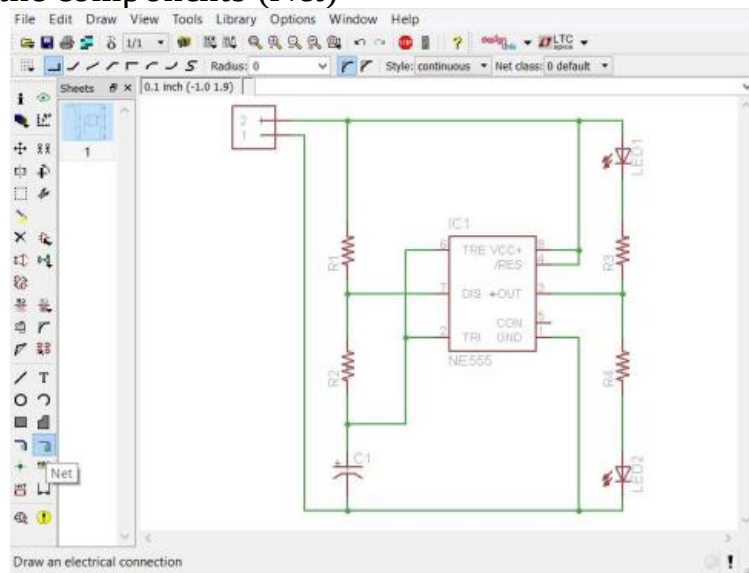
6. Add Component – LED and a Screw Terminal for Power Supply

- Adafruit: LED5MM (search “LED5mm”)
- Adafruit: 1X2-3.5MM (search “terminal”)

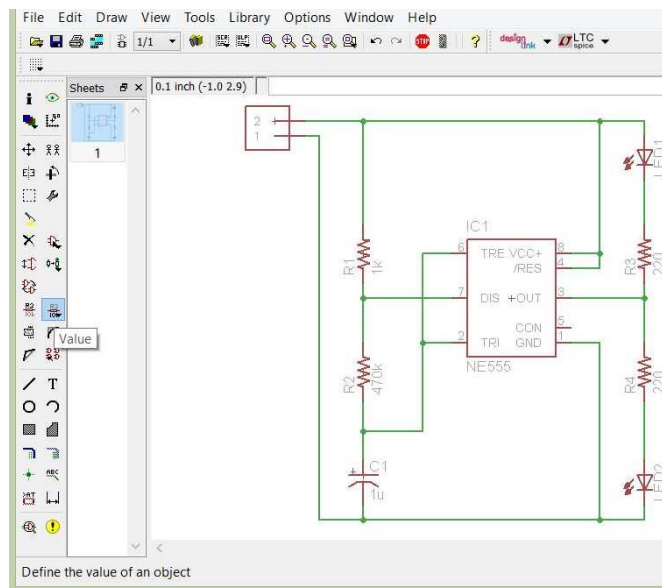
7. Rotating and Moving Components



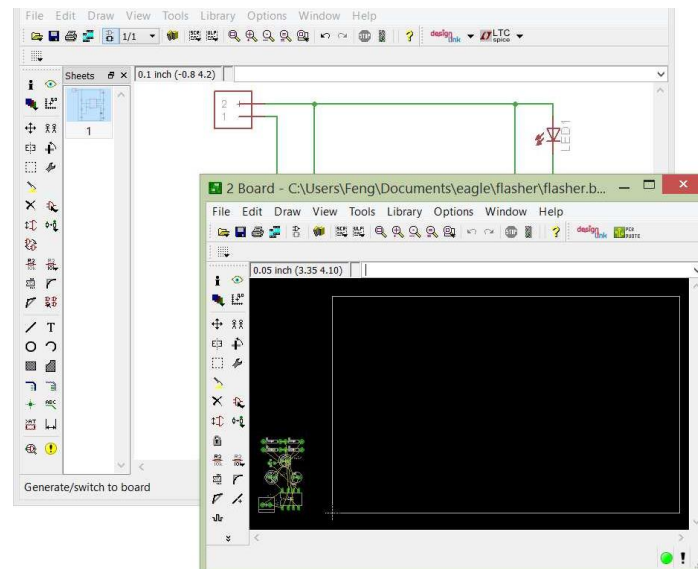
8. Connecting the components (Net)



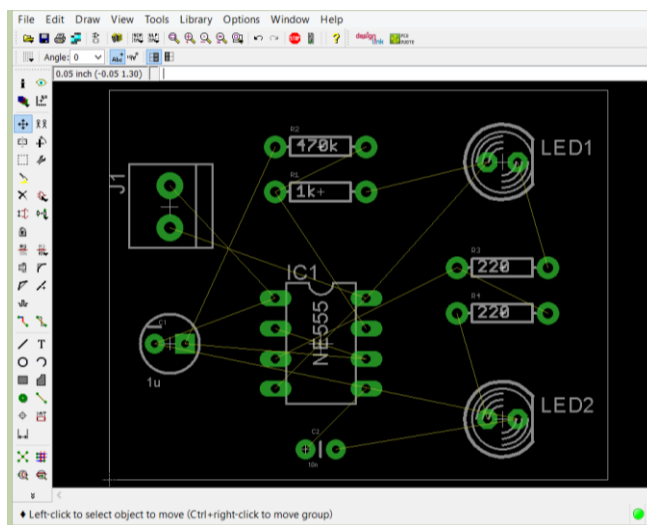
9. Setting Component Values



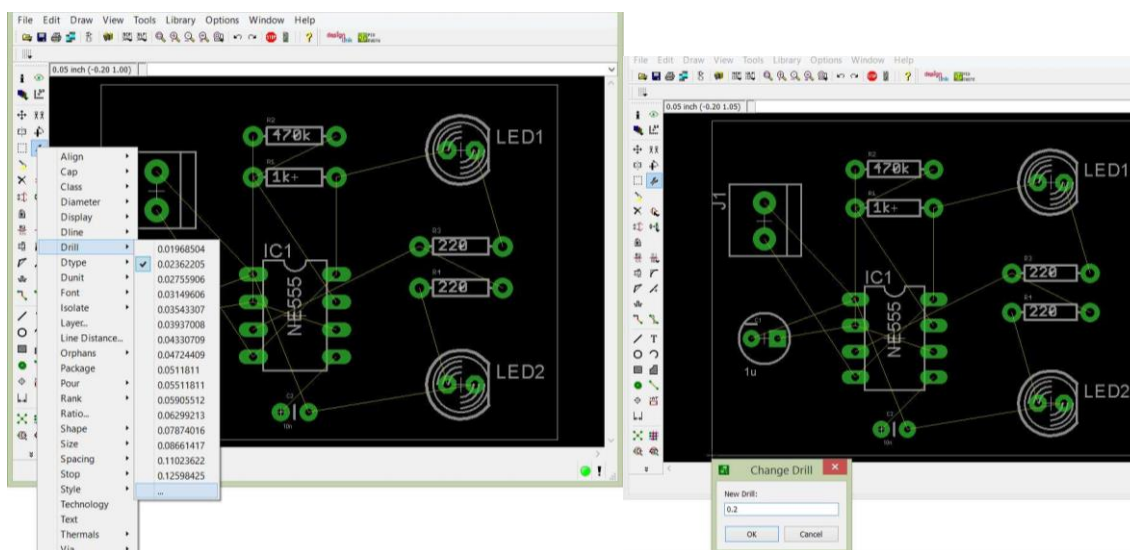
10. Laying Out the Board



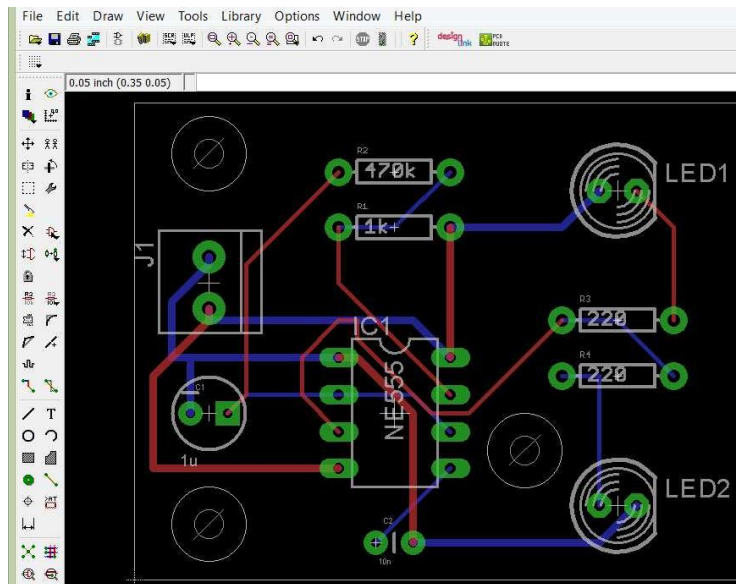
11. Move, rotate (resizing the board) Zoom to fit



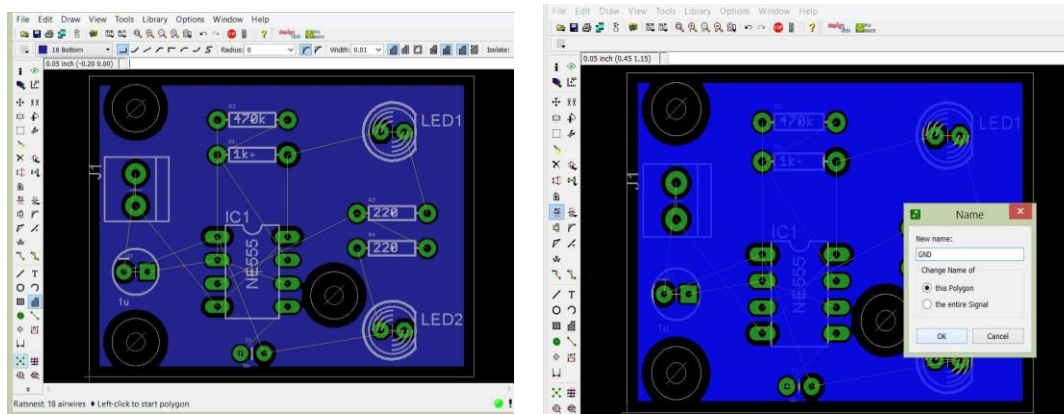
12.Add Mounting Holes and Autorouter



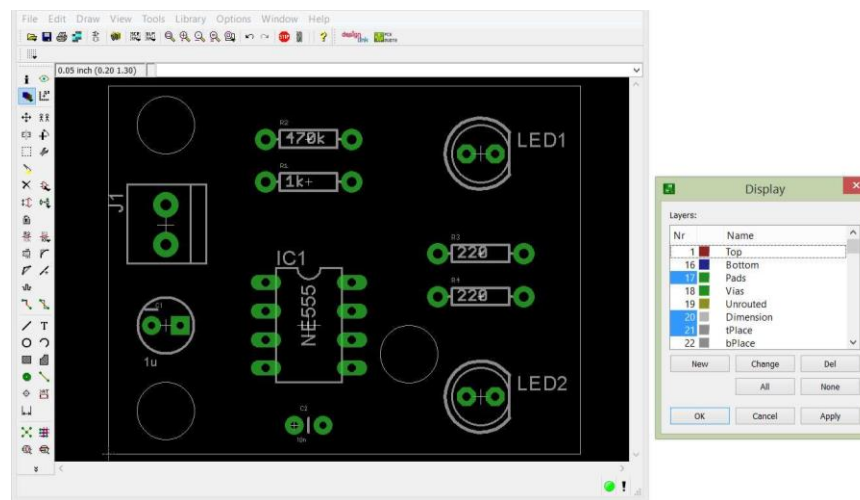
13.Board Layout



14. Add a Ground Plane



15. Add Text on the Top (Silkscreen)



16.Final layout board

